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Transient Control of Winding Reconfiguration for PMSMs Based on Deadbeat Predictive Control and Zero-Vector Switching

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Abstract

Stator winding reconfiguration technology has attracted increasing attention because it can modify the torque–speed output characteristics of motors online and extend the high-efficiency operating region. However, conventional winding switching methods suffer from several limitations, including long transient duration, typically at the millisecond level; difficulty in precise control; current interruption or asymmetric operation during the transient process; and the resulting torque loss and torque fluctuation. To address these issues, this paper first proposes a seamless winding switching method. The core idea is to complete the switching within the zero-vector interval of a single SVPWM period, thereby eliminating potential surge voltage without requiring a snubber circuit. The proposed method reduces the transient duration to only 5 μs, eliminates current interruption and current asymmetry, and suppresses the associated torque ripple. Furthermore, a deadbeat predictive control strategy is employed. By directly controlling the voltage vector, the proposed strategy enables fast and accurate current tracking within one sampling period after reconfiguration, thereby avoiding current overshoot, oscillation, and the corresponding torque and speed fluctuations. In addition, a voltage feedforward compensation method is proposed to pre-compensate the voltage applied during the PWM period in which the winding reconfiguration is performed, thereby preventing the application of voltage vectors that are inconsistent with the actual motor winding connection. With these three innovative measures, fast, smooth, and robust winding reconfiguration is achieved. The mechanisms and implementation procedures of the proposed strategies are described in detail, and their feasibility and effectiveness are verified through experiments.

Keywords: permanent magnet synchronous motor (PMSM); winding reconfiguration; seamless switching; zero-voltage vector (ZVV) interval; deadbeat predictive current control (DPCC)

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1. Introduction

To address global environmental challenges and the energy crisis, electric vehicles (EVs) are undergoing rapid development [1,2]. At present, most EVs adopt a drive topology consisting of a permanent magnet synchronous motor (PMSM) and a single-speed reducer. However, under certain operating conditions, it is difficult for this configuration to simultaneously satisfy the requirements of high torque at low speed and high power at high speed. Stator winding reconfiguration has emerged as a promising solution, as it can change the connection mode of the stator windings through external circuits [3]. This technique enables dynamic adjustment of the motor torque–speed output characteristics and effectively extends the high-efficiency operating region of the motor while providing performance similar to that of a conventional multi-speed transmission [4].

According to the differences in winding connection topologies, winding reconfiguration can be classified into several types, including star–delta reconfiguration [5–7], tapped-winding reconfiguration [8–10], series–parallel reconfiguration [11–14], and phase-shift reconfiguration [15–17]. In addition, hybrid reconfiguration schemes combining star–delta reconfiguration with other methods have also been reported in the literature [18–20]. Among these alternatives, series–parallel reconfiguration has received the broadest research attention because of its remarkable transformation ratio, continuous utilization of all windings, and relatively simple implementation process.

The switching devices used in the reconfiguration circuit can be mechanical components such as relays or contactors [11,21,22]. Although these devices have the advantages of low cost and low conduction loss, they also suffer from slow switching speed, large volume, and limited life. Therefore, power electronic switches, including IGBTs, MOSFETs, and thyristors, have become the preferred choice for reconfigurable winding systems [7–10].

The major challenge in winding reconfiguration is how to safely change the winding connections during motor operation. Sudden switching can generate surge voltage due to abrupt changes in the inductive current, which may damage the reconfiguration switches or even the inverter. Although integrating dedicated snubber circuits [23,24] provides an effective countermeasure, it inevitably increases the system volume, weight, and cost. Currently, the most widely used methods are the zero-current method (ZC) [11,22] and the phase-by-phase method (PBP) [13,25].

In the ZC method, the three-phase currents must first be reduced to zero, and the current is then re-established after winding reconfiguration. Although this method ensures the safety of the reconfiguration process, it introduces a new problem: the torque is completely interrupted, resulting in a consequent speed drop. The PBP method avoids torque interruption by changing the winding connection sequentially at the zero-crossing instants of each phase current. However, during this process, the motor operates under an asymmetric condition, which still causes torque fluctuation. Moreover, this process requires at least one third of an electrical current period.

The transition duration of these two methods is usually at the millisecond level and is affected by multiple factors. For the ZC method, the influencing factors include the inductance value, current amplitude, and current sampling noise. For the PBP method, the key influencing factor is the motor speed. In addition, because thyristors have natural commutation characteristics, the PBP method is generally more suitable for applications in which thyristors are used as reconfiguration switches. Otherwise, inaccurate zero-crossing detection may still lead to surge voltage generation.

For EV applications, when fully-controlled fast-switching power devices such as IGBTs, MOSFETs, or even SiC MOSFETs are used as reconfiguration switches [7,9,10,17,25], it becomes particularly important to develop a high-performance switching control strategy that is well matched with the hardware.

Deadbeat predictive current control (DPCC) has been widely applied in motor drive control systems because of its extremely fast dynamic response and excellent steady-state performance [26,27]. By calculating the required voltage vector applied to the motor, DPCC can achieve tracking of the reference current within a single control period.

This paper proposes an integrated control strategy for series–parallel winding reconfiguration of PMSM using fully-controlled power electronic devices as reconfiguration switches. The proposed strategy combines a seamless switching method with a DPCC scheme that incorporates feedforward voltage compensation.

The main contributions of this study are summarized as follows.

A seamless switching method based on the zero-voltage-vector (ZVV) interval is proposed. This method eliminates the potential surge voltage, torque interruption, and asymmetric operation problems associated with conventional switching methods. Meanwhile, the transient duration of winding reconfiguration is reduced to a fixed microsecond-level timescale, which is 5 μs in this study.

DPCC is introduced into the winding reconfiguration control system for the first time, and a feedforward voltage compensation method is proposed. Combined with the proposed seamless switching method, the strategy enables fast and accurate tracking of the step-changed current command within one control period, thereby effectively suppressing current overshoot and oscillation after reconfiguration.

The remainder of this paper is organized as follows. Section 2 introduces the series–parallel winding reconfiguration system. Section 3 presents the principles and implementation details of the proposed strategy. Section 4 shows the experimental results. Section 5 compares and discusses the characteristics and essential differences between the existing winding-reconfiguration control schemes and the proposed method. Finally, Section 6 concludes the paper.

2. The Series–Parallel Winding Reconfiguration System

Figure 1a shows the topology of the proposed system, which mainly consists of an inverter, a winding reconfiguration circuit, and a reconfigurable PMSM. As shown in Figure 1b, the PMSM contains two independent sets of three-phase windings, namely A1/B1/C1 and A2/B2/C2, whose magnetic axes are spatially aligned. The winding reconfiguration circuit consists of nine reconfiguration switches, including series switches S1–S3 and parallel switches P1–P6. The structure of a single switch is shown in the dashed box in Figure 1a, where a common-emitter IGBT module with integrated freewheeling diodes is adopted. When the motor operates in the series mode, S1–S3 are turned on and P1–P6 are turned off. When the motor is switched to the parallel mode, P1–P6 are turned on and S1–S3 are turned off.

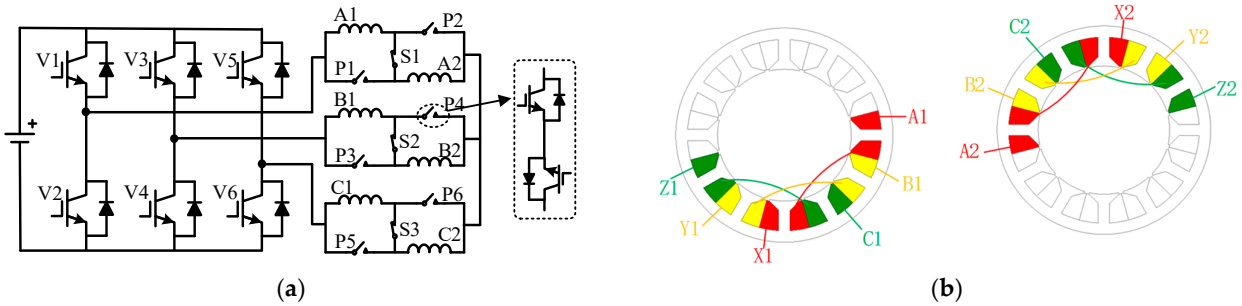


Figure 1. Series–parallel winding reconfiguration system: (**a**) system topology; (**b**) schematic diagram of the winding arrangement.

In the low-speed region, the series mode is adopted to obtain a higher rated torque. After the speed exceeds the base speed, switching to the parallel mode can double the constant-torque operating range. Figure 2 qualitatively illustrates the speed–torque output characteristics of the reconfigurable PMSM under different operating modes. Table 1 lists the parameter relationships under different winding connection modes. Under constant dc-link voltage and inverter current limit, the torque output capability or speed operating range of the reconfigurable PMSM can reach twice that of a conventional motor. In other words, if the rated power remains unchanged, the required volume and weight of the PMSM can be reduced accordingly.

Table 1. Parameter relationships under different winding modes.

Item	Resistance (p.u.)	Inductance (p.u.)	Flux (p.u.)	Rated Torque (p.u.)	Base Speed (p.u.)
Series mode	1	1	1	1	1
Parallel mode	0.25	0.25	0.5	0.5	2

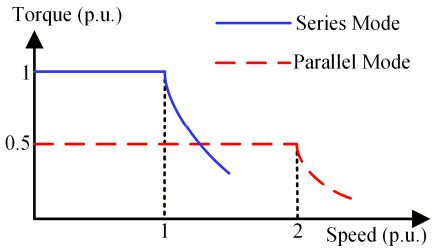


Figure 2. Qualitative speed–torque output characteristics of the reconfigurable PMSM.

3. The Proposed Strategy

3.1. The Winding Switching Method Based on the ZVV Interval

3.1.1. Basic Principle

A two-level voltage source inverter can generate six active voltage vectors, $\boldsymbol{u}_1$–$\boldsymbol{u}_6$, and two ZVVs, $\boldsymbol{u}_0$ and $\boldsymbol{u}_7$, forming the vector space shown in Figure 3. In Figure 3, the orange arrows denote the $\alpha\beta$ coordinate system, the black arrows represent the basic voltage vectors, and the blue arrows stand for the synthesized voltage vectors. According to the minimum switching action principle, Taking Sector I as an example, Figure 4 shows the switching sequence of seven-segment space vector pulse-width modulation (SVPWM). In each SVPWM period, the ZVVs are distributed at the beginning, middle, and end of this period.

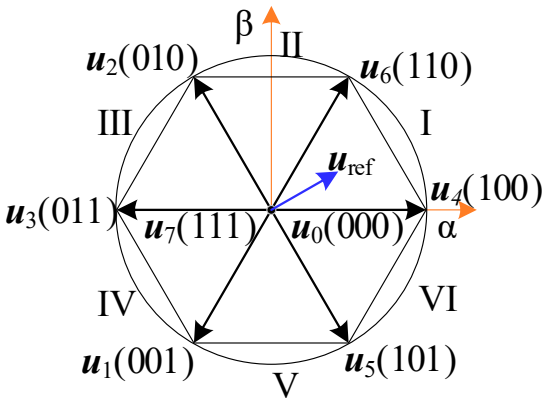


Figure 3. Vector space.

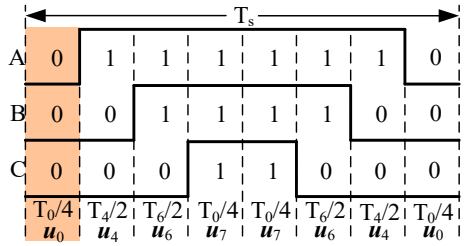


Figure 4. SVPWM switching sequence.

The proposed seamless switching method based on the ZVV interval refers to performing winding reconfiguration during the interval in which $\boldsymbol{u}_0$ or $\boldsymbol{u}_7$ is applied. If winding reconfiguration is performed during the $\boldsymbol{u}_7$ interval, there will be at least half a sampling period during which the applied voltage vector does not match the winding connection state of the motor. Therefore, in this study, winding reconfiguration is performed during the $\boldsymbol{u}_0$ interval at the beginning of one SVPWM period, as indicated by the orange region in Figure 4.

3.1.2. Mode Decomposition of the Reconfiguration Process

During winding reconfiguration, the operating states of the system can be divided into three modes: series mode, transient mode, and parallel mode. Taking the case in which the reference ZVV is located in Sector I and the motor is switched from the series mode to the parallel mode as an example, the operating states of the system in the three modes are described in detail below.

Mode 1: As shown in Figure 5, the windings operate in the series mode, where the series switches S1–S3 are turned on and the parallel switches P1–P6 are turned off. At the end of the previous SVPWM period, the output voltage vector of the inverter switches from $\boldsymbol{u}_4$ to $\boldsymbol{u}_0$. The current freewheels through the lower bridge arms of the inverter, the motor windings, and switches S1–S3. The current flow direction is indicated by the dashed arrows. This state continues into the $\boldsymbol{u}_0$ interval at the beginning of the current SVPWM period.

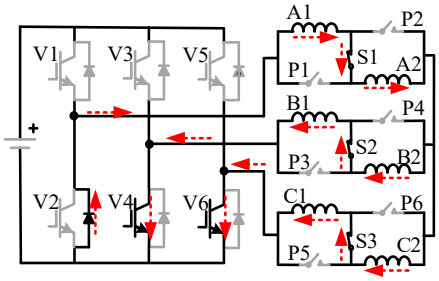


Figure 5. Mode 1—The series mode.

Mode 2: As shown in Figure 6, the states of the inverter and switches S1–S3 remain the same as those in Mode 1, while the parallel switches P1–P6 are switched from off to on. This is the transient mode, in which all nine reconfiguration switches are turned on. The current freewheels through the lower bridge arms of the inverter, the motor windings, and all reconfiguration switches. Although the freewheeling loop changes, the magnitude and direction of the current in each winding remain unchanged. Therefore, theoretically, no surge voltage or abrupt torque change is generated.

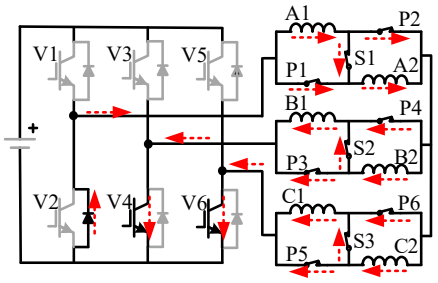


Figure 6. Mode 2—The transient mode.

Mode 3: As shown in Figure 7, the states of the inverter and switches P1–P6 remain the same as those in Mode 2, while the series switches S1–S3 are turned off. The current freewheels through the lower bridge arms of the inverter, the motor windings, and switches P1–P6. At this point, the winding reconfiguration process is completed, and the motor enters the parallel operating mode. The inverter output voltage remains at $\boldsymbol{u}_0$ until the active voltage vector $\boldsymbol{u}_4$ is applied.

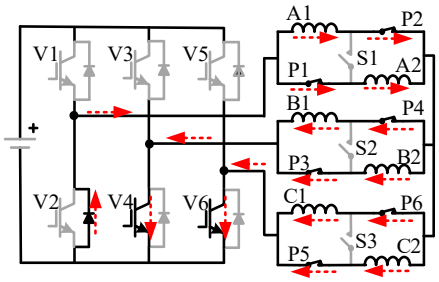


Figure 7. Mode 3—The parallel mode.

Table 2 lists the ON/OFF states of each reconfiguration switch under different operating modes.

Table 2. Conduction states of reconfiguration switches.

Item	S1	S2	S3	P1	P2	P3	P4	P5	P6
Mode 1	1	1	1	0	0	0	0	0	0
Mode 2	1	1	1	1	1	1	1	1	1
Mode 3	0	0	0	1	1	1	1	1	1

3.1.3. Determination of the Minimum ZVV Interval

When winding reconfiguration is performed, the duration of the ZVV interval at the beginning of the SVPWM period should be no less than the winding reconfiguration transient time, T_{tr}.

This transient time should include the turn-on time (t_{on}) and the turn-off time (t_{off}) of the reconfiguration switches, as shown in (1). These parameters can be obtained from the device datasheet, while the necessary signal propagation delay and safety margin should also be considered. In this work, t_{on} = 3 μs and t_{off} = 2 μs are adopted; therefore, the minimum ZVV interval is set as T_{tr} = 5 μs. If the motor operates under a high modulation index, resulting in an insufficient ZVV duration, it can be directly limited to 5 μs in that specific SVPWM period. Since this situation occurs with a very low frequency and affects only one SVPWM period each time, its negative influence on system performance can be neglected.

The decomposition of the reconfiguration transient time is shown in Figure 8. From top to bottom, the figure shows the phase-A switching sequence, which has been presented in Figure 4, and the PWM waveforms of the upper and lower inverter bridge arms of phase A.

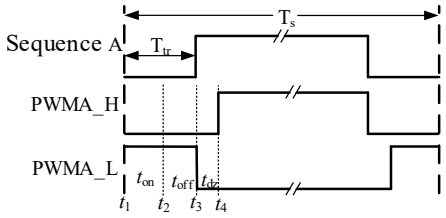


Figure 8. Decomposition of the reconfiguration transient time.

The meanings of the time instants are explained as follows. The instant t_1 is the starting instant of this SVPWM period, that is, the starting instant of the ZVV interval $\boldsymbol{u}_0$. At this instant, turn-on signals are synchronously applied to switches P1–P6. By the instant t_2, P1–P6 have been fully turned on, and at this instant, turn-off signals are synchronously sent to switches S1–S3. The instant t_3 represents the end of the ZVV interval, and the inverter starts to output the active voltage vector $\boldsymbol{u}_4$. At this instant, S1–S3 have been completely turned off, the reconfiguration transient process is also terminated accordingly, and the motor enters the parallel winding mode. The interval between t_3 and t_4 represents the dead time of the PWM signal.

$$T_{tr} = t_{on} + t_{off}, \tag{1}$$

3.2. Control Strategy Incorporating DPCC

3.2.1. Principle of DPCC

This study focuses on a surface-mounted PMSM. Its mathematical model in the dq reference frame can be expressed as:

$$\begin{cases} \dfrac{di_d}{dt} = \dfrac{1}{L}u_d - \dfrac{R}{L}i_d + \omega_e i_q \\ \dfrac{di_q}{dt} = \dfrac{1}{L}u_d - \dfrac{R}{L}i_q - \dfrac{1}{L}\omega_e(Li_d + \psi_f) \end{cases} \tag{2}$$

where u and i denote the stator voltage and current, respectively; the subscripts d and q represent the d-axis and q-axis components, respectively; L is the stator inductance; R is the stator resistance; ω_e is the rotor electrical angular speed; and ψ_f is the permanent magnet flux linkage.

Based on (2), forward Euler discretization is performed, and the predicted current expressions for the next sampling period can be obtained as follows:

$$\begin{cases} i_{dp}^{k+1} = \left(1 - \dfrac{T_s R}{L}\right) i_d^k + T_s \omega_e i_q^k + \dfrac{T_s}{L} u_d^k \\ i_{qp}^{k+1} = \left(1 - \dfrac{T_s R}{L}\right) i_q^k - T_s \omega_e i_d^k + \dfrac{T_s}{L} u_q^k - \dfrac{T_s}{L}\omega_e \psi_f \end{cases} \tag{3}$$

where i_{dp}^{k+1} and i_{qp}^{k+1} are the predicted currents in the (k + 1)th sampling period; i_d^k and i_q^k are the actual currents in the (k)th sampling period; u_d^k and u_q^k are the applied voltage vector; and T_s is the sampling period.

The objective of DPCC is to ensure that the currents at instant (k + 1) are infinitely close to the reference values at instant k, thereby eliminating the tracking error. Therefore,

it is essential to calculate the most appropriate voltage vector. Based on the discrete mathematical model in (3), the stator voltage at instant k can be obtained as:

$$\begin{cases} u_{\mathrm{d}}^{k} = \dfrac{L}{T_{\mathrm{s}}} i_{\mathrm{d}}^{\mathrm{ref}} + \left(R - \dfrac{L}{T_{\mathrm{s}}} \right) i_{\mathrm{d}}^{k} - \omega_{\mathrm{e}} L i_{\mathrm{q}}^{k} \\ u_{\mathrm{q}}^{k} = \dfrac{L}{T_{\mathrm{s}}} i_{\mathrm{q}}^{\mathrm{ref}} + \left(R - \dfrac{L}{T_{\mathrm{s}}} \right) i_{\mathrm{q}}^{k} + \omega_{\mathrm{e}} L i_{\mathrm{d}}^{k} + \omega_{\mathrm{e}} \psi_{\mathrm{f}} \end{cases} \tag{4}$$

where i_{d}^{k+1} and i_{q}^{k+1} are replaced by their reference values $i_{\mathrm{d}}^{\mathrm{ref}}$ and $i_{\mathrm{q}}^{\mathrm{ref}}$. respectively.

Considering the one-step delay in the practical system, the predicted stator voltage in the (k + 1)th sampling period can be expressed as:

$$\begin{cases} u_{\mathrm{d}}^{k+1} = \dfrac{L}{T_{\mathrm{s}}} i_{\mathrm{d}}^{\mathrm{ref}} + \left(R - \dfrac{L}{T_{\mathrm{s}}} \right) i_{\mathrm{dp}}^{k+1} - \omega_{\mathrm{e}} L i_{\mathrm{qp}}^{k+1} \\ u_{\mathrm{q}}^{k+1} = \dfrac{L}{T_{\mathrm{s}}} i_{\mathrm{q}}^{\mathrm{ref}} + \left(R - \dfrac{L}{T_{\mathrm{s}}} \right) i_{\mathrm{qp}}^{k+1} + \omega_{\mathrm{e}} L i_{\mathrm{dp}}^{k+1} + \omega_{\mathrm{e}} \psi_{\mathrm{f}} \end{cases} \tag{5}$$

where i_{dp}^{k+1} and i_{qp}^{k+1} can be calculated using (3).

In the winding reconfiguration control system, motor parameters, current references and actual phase currents may all experience step changes. If conventional DPCC is directly adopted for reconfiguration transient control without dedicated modifications, torque oscillations or even control instability will occur after reconfiguration. Accordingly, a feedforward voltage compensation measure is proposed for DPCC targeting winding reconfiguration transient control.

3.2.2. Implementation of DPCC with Voltage Compensation During Winding Reconfiguration

When the motor windings operate in the series steady state or the parallel steady state, the DPCC algorithm is implemented in the same manner as in conventional operation, namely according to (3) and (5). The motor parameters corresponding to the series mode are used in the series steady state, while those corresponding to the parallel mode are used in the parallel steady state. When winding reconfiguration occurs, several special treatments are required after the reconfiguration command is received.

Taking the series-to-parallel winding reconfiguration as an example, Figure 9, together with (6) and (7), illustrates the implementation of DPCC during the reconfiguration process. Assuming that the reconfiguration command is updated in the (k)th sampling period, the following operations should be performed in this period: (1) the motor parameters and control parameters are changed to those of the parallel mode; (2) the voltage vector with feedforward compensation, u_{dq}^{k+1}, for the next sampling period is predicted; and (3) the length of the ZVV interval calculated for the (k + 1)th period is checked to determine whether it satisfies the required transient time T_{tr}. If not, it is limited to T_{tr}. At the beginning of the next sampling period, namely the (k + 1)th period, the winding reconfiguration action can be executed.

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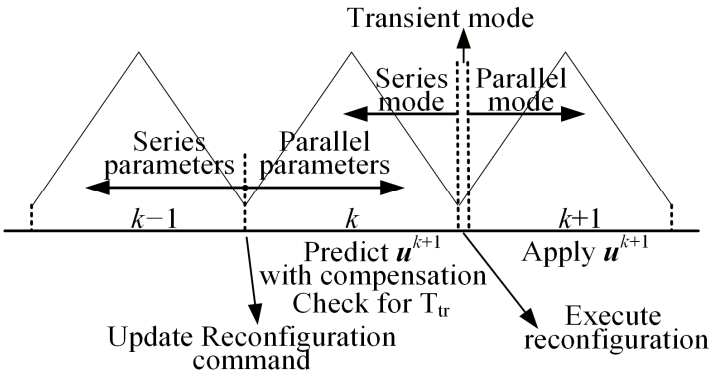


Figure 9. Implementation of the predictive algorithm with voltage compensation for winding reconfiguration (Series to Parallel).

The reasons for adopting feedforward compensation on u_{dq}^{k+1} and the detailed calculation method are elaborated as follows:

First, in the (k)th period, the motor windings are still operating in the series mode. Therefore, the series motor parameters should still be used to calculate the predicted current i_{dq}^{k+1} in (3), which can be rewritten as (6). For u_{dq}^{k+1}, however, since it will be applied in the (k + 1)th sampling period, at which time the winding reconfiguration has already been completed, the parallel motor parameters should be used for its calculation. Furthermore, because the motor has already entered the parallel mode, the actual phase current in the (k + 1)th period is twice the predicted value i_{dp}^{k+1} and i_{qp}^{k+1}. Therefore, it should be multiplied by 2, as shown in (7).

$$\begin{cases} i_{dp}^{k+1} = \left(1 - \dfrac{T_s R_S}{L_S}\right) i_d^k + T_s \omega_e i_q^k + \dfrac{T_s}{L_S} u_d^k \\ i_{qp}^{k+1} = \left(1 - \dfrac{T_s R_S}{L_S}\right) i_q^k - T_s \omega_e i_d^k + \dfrac{T_s}{L_S} u_q^k - \dfrac{T_s}{L_S} \omega_e \psi_{fS} \end{cases} \tag{6}$$

$$\begin{cases} u_d^{k+1} = \dfrac{L_P}{T_s} i_d^{ref} + \left(R_P - \dfrac{L_P}{T_s}\right)\left(2i_{dp}^{k+1}\right) - \omega_e L_P \left(2i_{qp}^{k+1}\right) \\ u_q^{k+1} = \dfrac{L_P}{T_s} i_q^{ref} + \left(R_P - \dfrac{L_P}{T_s}\right)\left(2i_{qp}^{k+1}\right) + \omega_e L_P \left(2i_{dp}^{k+1}\right) + \omega_e \psi_{fP} \end{cases} \tag{7}$$

The capital subscripts S and P of the motor parameters denote the series and parallel parameters, respectively.

Similarly, for reconfiguration from the parallel mode to the series mode, the calculation in the (k)th sampling period is performed according to (8) and (9).

$$\begin{cases} i_{dp}^{k+1} = \left(1 - \dfrac{T_s R_P}{L_P}\right) i_d^k + T_s \omega_e i_q^k + \dfrac{T_s}{L_P} u_d^k \\ i_{qp}^{k+1} = \left(1 - \dfrac{T_s R_P}{L_P}\right) i_q^k - T_s \omega_e i_d^k + \dfrac{T_s}{L_P} u_q^k - \dfrac{T_s}{L_P} \omega_e \psi_{fP} \end{cases} \tag{8}$$

$$\begin{cases} u_d^{k+1} = \dfrac{L_S}{T_s} i_d^{ref} + \left(R_S - \dfrac{L_S}{T_s}\right)\left(0.5i_{dp}^{k+1}\right) - \omega_e L_S \left(0.5i_{qp}^{k+1}\right) \\ u_q^{k+1} = \dfrac{L_S}{T_s} i_q^{ref} + \left(R_S - \dfrac{L_S}{T_s}\right)\left(0.5i_{qp}^{k+1}\right) + \omega_e L_S \left(0.5i_{dp}^{k+1}\right) + \omega_e \psi_{fS} \end{cases} \tag{9}$$

It should be emphasized that the feedforward voltage compensation is required only once in the specific SVPWM period corresponding to winding reconfiguration. Therefore, it introduces almost no additional computational burden to the control algorithm.

Figure 10 presents a flowchart of the motor winding reconfiguration process from the series mode to the parallel mode.

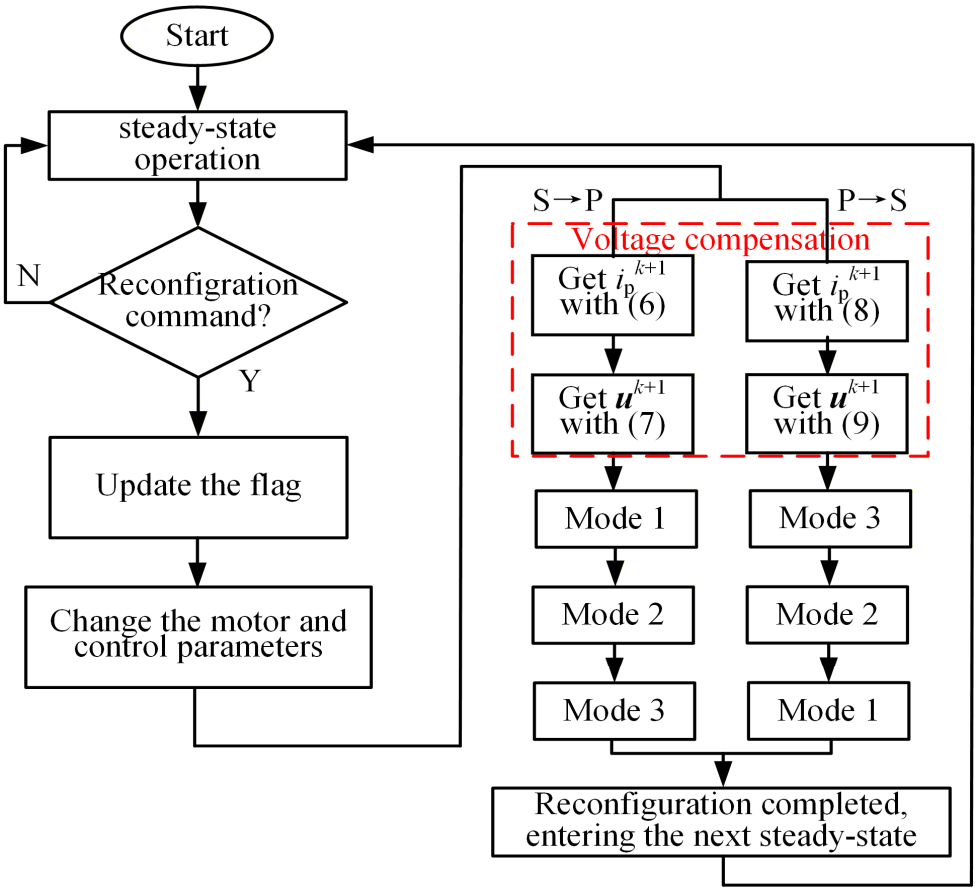


Figure 10. Implementation flowchart.

3.3. Overview of the Drive and Control System

Figure 11 shows the block diagram of the winding-reconfigurable motor drive and control system investigated in this study. The speed loop adopts conventional PI control, while the current loop adopts DPCC. In the reconfiguration algorithm, the parameters that need to be updated according to the winding reconfiguration command include the motor parameters, namely resistance, inductance, and flux linkage, and the control parameters, namely the speed-loop PI parameters and the q-axis current gain. Theoretically, the speed-loop PI parameters in the series mode are 0.5 times those in the parallel mode [28]. Since the output of the speed-loop integrator cannot change in a stepwise manner, a q-axis current gain selector is introduced in the current loop to rapidly change the current command after winding reconfiguration.

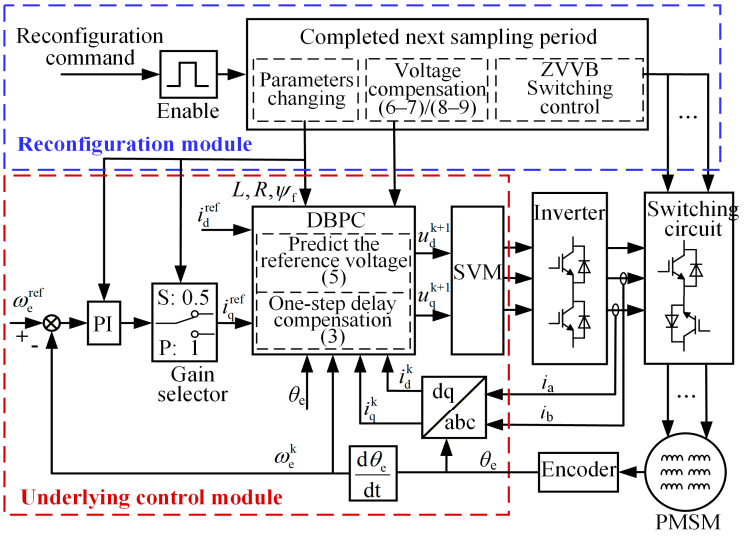


Figure 11. System block diagram.

4. Experimental Validation

To verify the performance of the proposed control strategy, a surface-mounted dual-winding PMSM is modified into a winding-reconfigurable topology, and the experimental platform shown in Figure 12 is established. The controller adopts a TMS320F28379D DSP. The reconfiguration switches and inverter power module are Infineon IKQB120N75CP2 and FF600R12ME4 devices, respectively. The detailed motor parameters are listed in Table 3. The dc-link voltage of the system is set to 24 V, and the PWM frequency is 10 kHz.

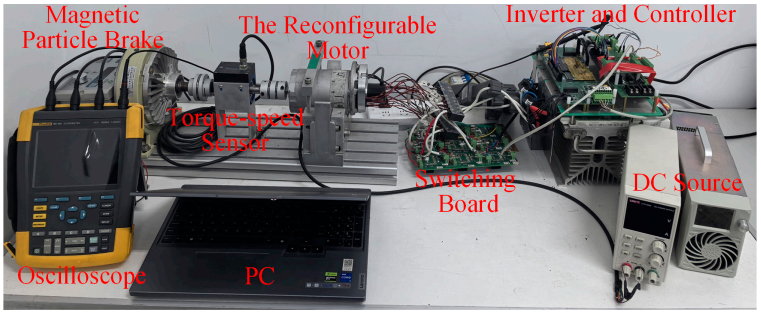


Figure 12. The experimental platform.

In the following experimental results, the current waveforms are directly acquired using current clamps. The speed and torque waveforms cannot be measured directly. The speed is calculated based on the position information detected by a 2500-line encoder. The torque is derived from the phase currents measured by Hall effect current sensors with a sampling accuracy of 2%. These data are transmitted from the DSP to the host computer via serial communication and subsequently plotted as waveforms using Origin v2021. The uploading and updating frequency are identical to the current-loop sampling frequency (10 kHz).

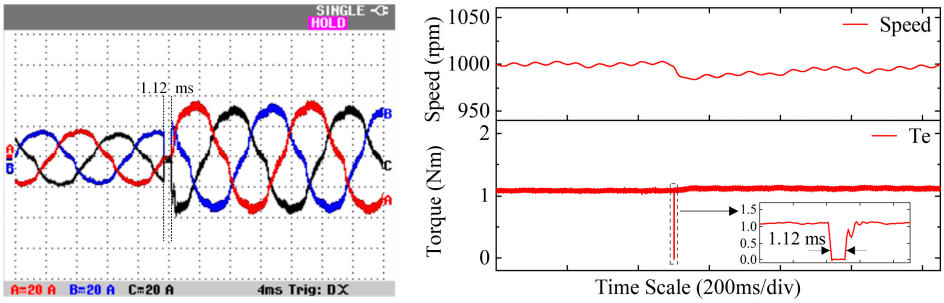
Table 3. Parameters of the motor.

Parameter	Parallel Mode	Series Mode
Stator resistance per phase (Ω)	5×10^{-3}	2×10^{-2}
Stator inductance per phase (H)	1.73×10^{-5}	6.92×10^{-5}
Permanent magnet flux (Wb)	5.67×10^{-3}	1.134×10^{-2}

Torque constant (Nm/Arms)	0.045	0.09
Number of pole pairs	4	4
Rated torque (Nm)	1	2
Rated speed (rpm)	4000	2000

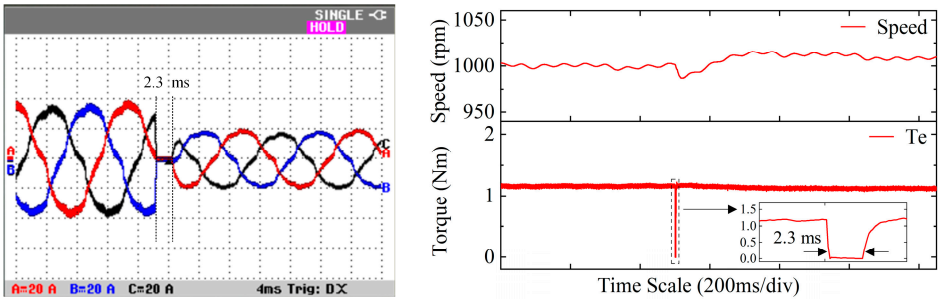
To verify the effectiveness of the proposed winding reconfiguration transient control strategy, namely the ZVV-based DPCC (ZVV-DPCC), comparative experiments are conducted against the widely used zero-current (ZC) method. For the ZC method, the current loop is implemented using field-oriented control (FOC). In addition, a q-axis current gain selector is also configured to enable step adjustment of the current command after winding reconfiguration is completed.

First, winding reconfiguration is performed under constant-speed and constant-torque operating conditions, with a speed of 1000 rpm and a torque of 1 N·m. Figures 13 and 14 show the experimental results of the ZC method. When receiving the reconfiguration command, the current reference is immediately set to zero, and the three-phase currents decay rapidly. The triggering condition for reconfiguration in this method is that the instantaneous values of the three-phase currents are simultaneously lower than 0.5 A.



(**a**) (**b**)

Figure 13. Experimental results of ZC during series-to-parallel reconfiguration: (**a**) Phase currents. (**b**) Speed and torque.



(**a**) (**b**)

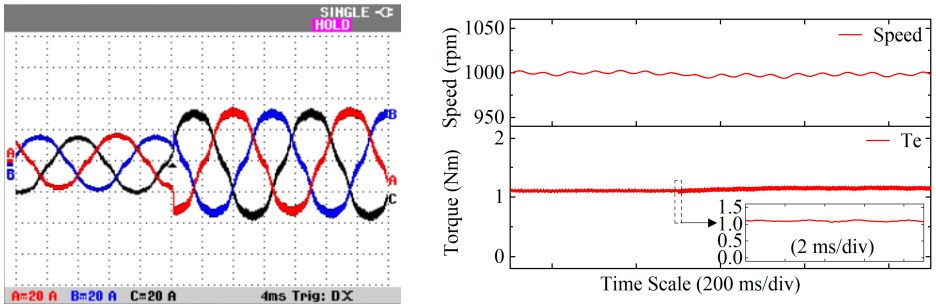
Figure 14. Experimental results of ZC during parallel-to-series reconfiguration: (**a**) Phase currents. (**b**) Speed and torque.

Figure 13a,b show the three-phase current, speed, and torque waveforms of the ZC method during series-to-parallel reconfiguration. It can be observed that although the three-phase currents decrease to nearly zero within three current sampling periods, the actual reconfiguration action is delayed by 1.12 ms. This delay is caused by current sampling noise, which prevents the triggering condition from being satisfied promptly. During this process, the three-phase currents are interrupted, resulting in complete torque interruption and a subsequent speed drop. The torque interruption lasts for 1.12 ms, and the speed drop is 18 rpm.

Figure 14a,b show the corresponding waveforms under the parallel-to-series reconfiguration condition. The characteristics are similar to those described above. Under this operating condition, the torque interruption lasts for 2.3 ms, and the speed decreases by 25 rpm.

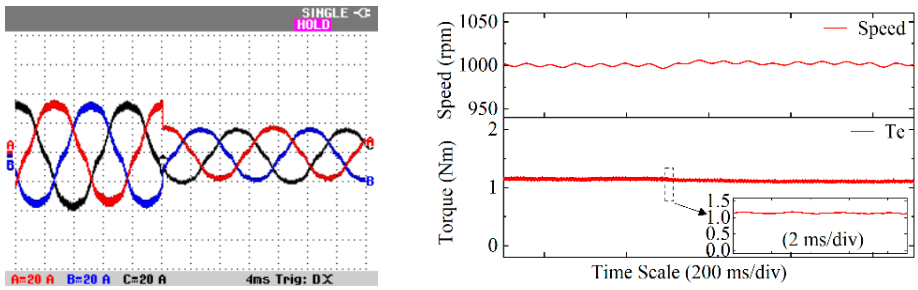
These results indicate that the ZC method inevitably produces an interruption interval in the three-phase currents. The resulting torque interruption and speed drop are the main causes of performance degradation in the winding reconfiguration system. The winding reconfiguration transient time is typically on the millisecond scale and is also affected by current sampling noise; therefore, it cannot be accurately predicted or controlled.

Figures 15 and 16 show the experimental results of the proposed ZVV-DPCC method. As analyzed in Section 3, the smooth switching method based on the ZVV interval can complete winding reconfiguration within the second sampling period after the reconfiguration command is received, while eliminating surge voltage and torque interruption. The duration of the transient process is fixed at 5 μs and is independent of the motor operating conditions. In addition, by combining DPCC with voltage feedforward compensation, fast and accurate current tracking can be achieved without current spikes, oscillations, or overshoot.



(**a**) (**b**)

Figure 15. Experimental results of ZVV-DPCC during series-to-parallel reconfiguration: (**a**) Phase currents. (**b**) Speed and torque.



(**a**) (**b**)

Figure 16. Experimental results of ZVV-DPCC during parallel-to-series reconfiguration: (**a**) Phase currents. (**b**) Speed and torque.

The experimental results verify the theoretical analysis. Under both the series-to-parallel reconfiguration (Figure 15) and the parallel-to-series reconfiguration (Figure 16), the torque remains stable without fluctuation, and the maximum speed deviation is only 5 rpm. Considering the steady-state speed fluctuation of ±3 rpm, it can be considered that the speed is also not affected by the reconfiguration transient. These results demonstrate the excellent control performance of the proposed strategy during the winding reconfiguration transient.

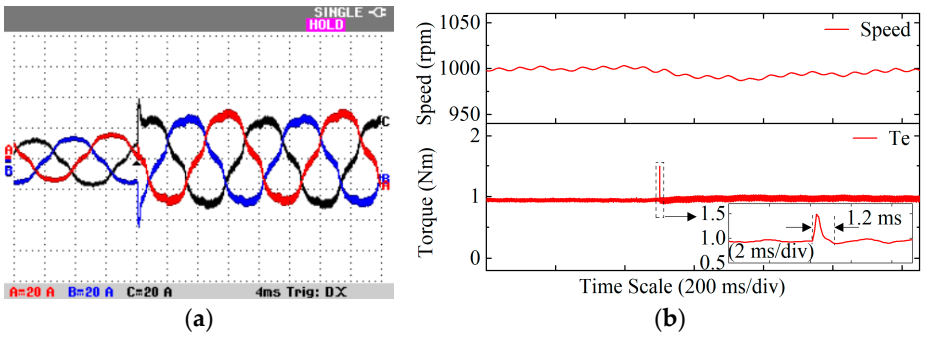
Table 4 summarizes the quantitative metrics of the ZC method and the ZVV-DPCC method during winding reconfiguration under constant-speed and constant-load conditions, including transient time, torque performance, and the maximum speed deviation relative to the speed reference.

Table 4. Quantitative metrics of ZC and ZVV-DPCC methods.

Methods	Process	Transient Duration	Torque Ripple	Speed Deviation
ZC	series-to-parallel	1.12 ms	100% interrupt	18 rpm
	parallel-to-series	2.3 ms	100% interrupt	25 rpm
ZVV-DPCC	series-to-parallel	5 μs	None	4 rpm
	parallel-to-series	5 μs	None	5 rpm

The above comparative results indicate that the proposed ZVV-DPCC method provides superior control performance. Without adding any snubber circuit, the proposed method can avoid abrupt changes in the inductive current and suppress surge voltage. Its transient response is fast, and the transient duration is accurately controllable. Moreover, nearly no torque or speed fluctuation occurs during the winding reconfiguration transient, indicating excellent operating smoothness. These advantages make the ZVV-DPCC method particularly suitable for high-performance drive systems that require both fast and smooth winding reconfiguration.

To independently verify the necessity of introducing DPCC into winding reconfiguration control, the proposed ZVV-based winding switching scheme is implemented in combination with the FOC current-loop control. Taking the winding reconfiguration from series to parallel as an example, the experimental results are presented in Figure 17. It can be observed that the overshoot and regulation lag of the PI controllers in the current loop lead to a torque overshoot of 49.1% with a duration of 1.2 ms. This indicates that although the proposed ZVV-based method can complete the winding reconfiguration operation within 5 μs, its full potential cannot be fully exploited without DPCC to achieve one-step current tracking.

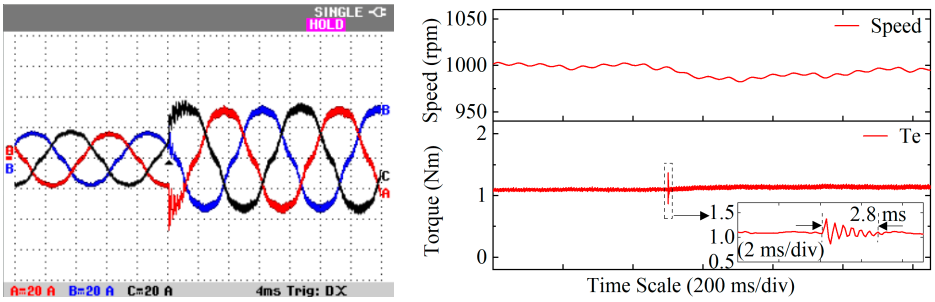


(**a**) (**b**)

Figure 17. Experimental results of the ZVV-based switching combined with FOC (series-to-parallel reconfiguration): (**a**) Phase currents. (**b**) Speed and torque.

Furthermore, to demonstrate the necessity of the proposed voltage compensation strategy for implementing DPCC during the reconfiguration transient, an experiment is conducted using ZVV-based switching integrated with DPCC but without voltage compensation. Again taking the series-to-parallel reconfiguration as an example, the experimental results are shown in Figure 18. As revealed in the theoretical analysis section, the absence of voltage compensation will result in voltage vectors that mismatch the actual winding connection state after reconfiguration, thereby inducing torque ripples. The experimental results verify the theoretical analysis. The torque ripples last for as long as 2.8

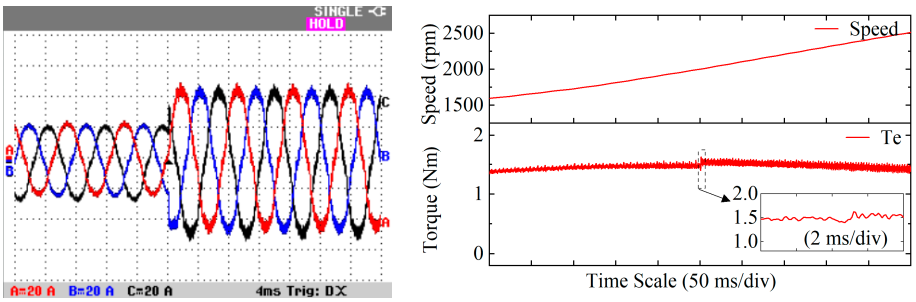
ms with a maximum ripple amplitude of 27.2%. Therefore, the proposed voltage compensation strategy is crucial for achieving smooth torque switching.



(**a**) (**b**)

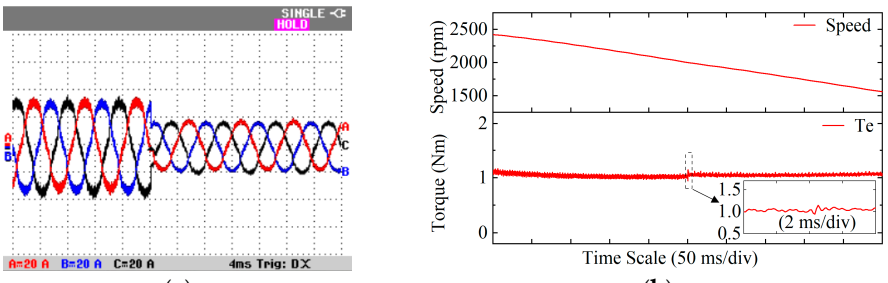
Figure 18. Experimental results of the ZVV-based switching combined with DPCC but without voltage compensation (series-to-parallel reconfiguration): (**a**) Phase currents. (**b**) Speed and torque.

Figures 19 and 20 present the experimental results of winding reconfiguration during the dynamic speed response. The operating condition of the motor is set as the rated load of the parallel mode with a speed variation rate of ±1500 rpm. Winding switching is automatically triggered when the actual speed accelerates or decelerates to the rated speed of the series mode. At this moment, the motor operates under the cross-rated operating condition between the parallel and series modes, which matches the practical triggering scenarios for winding reconfiguration. Figure 19 illustrates the reconfiguration process from series to parallel during acceleration, while Figure 20 depicts the reconfiguration process from parallel to series during deceleration. It can be observed that no obvious torque ripples or noticeable speed fluctuations occur during the reconfiguration transients in both cases, which demonstrates the adaptability of the proposed ZVV-DPCC scheme to dynamic operating conditions.



(**a**) (**b**)

Figure 19. Experimental results of ZVV-DPCC during series-to-parallel reconfiguration under acceleration conditions: (**a**) Phase currents. (**b**) Speed and torque.



(**a**) (**b**)

Figure 20. Experimental results of ZVV-DPCC during parallel-to-series reconfiguration under deceleration conditions: (**a**) Phase currents. (**b**) Speed and torque.

5. Discussion

The two conventional transient control methods most widely used for series-parallel winding reconfiguration are the ZC method [11,22] and the PBP method [13,25]. This section elaborates on the core characteristics and critical distinctions between these two classical approaches and the ZVV-DPCC scheme proposed in this work. Table 5 lists the respective characteristics of the three methods.

By comparison, the following conclusions can be drawn. The ZC method has the broadest applicability, as it is compatible with all types of reconfiguration switches. However, complete torque interruption and the consequent speed drop render it inadequate for high-performance electric drive systems. The PBP method is suitable for applications adopting thyristors as reconfiguration switches. It eliminates torque interruption yet introduces brief asymmetric operation, which still generates torque and speed fluctuations. Both the ZC and PBP methods feature millisecond-scale transient durations and are susceptible to motor operating conditions, sampling accuracy and other factors. The proposed ZVV-DPCC scheme is applicable to systems utilizing fully-controlled power electronic devices as reconfiguration switches. It achieves microsecond-level, precisely controllable transients and eliminates torque and speed fluctuations.

Table 5. Characteristics of the three transient control methods.

Item	ZC [11,22]	PBP [13,25]	Proposed ZVV-DPCC
Principle	Reduce three-phase currents to zero and reconstruct them after reconfiguration.	Carry out phase-by-phase reconfiguration at the zero-crossing point of each phase current.	Perform reconfiguration within the ZVV interval of a single SVPWM.
Applicability	All types of switches (mechanical, electronic).	Thyristors.	Fully-controlled electronic switches (IGBT, MOSFET, etc.).
Surge Voltage	None.	None.	None.
Transient Duration	Millisecond level; Exceeds 100 ms for mechanical switches.	Millisecond level, at least 1/3 of one current cycle.	Microsecond level; 5 μs in this work.
Factors Affecting Transient Duration	Reconfiguration switch types; Current amplitude; Time constant $\tau = L/R$; Current sampling noise.	Rotational speed; Current zero-crossing point detection accuracy.	Immune to external factors for a given switch.
Transient Feature	Complete current interruption.	Asymmetric operation.	No current interruption or asymmetric operation; only freewheeling path altered.
Torque Performance	Full interruption.	No interruption, but torque fluctuations occur during asymmetric operation.	No interruption or visible fluctuations.
Speed Performance	Inevitable speed drop.	Fluctuations exist during asymmetric operation.	Nearly no fluctuation.

6. Conclusions

This paper proposes an integrated control strategy for a series–parallel winding reconfiguration system using fully-controlled power electronic devices as reconfiguration switches. The proposed strategy combines a smooth winding switching method based on the ZVV interval with deadbeat predictive current control incorporating voltage feedforward compensation.

The proposed strategy effectively addresses a series of key issues associated with conventional winding switching methods, including long transient process, surge voltage, current interruption, torque and speed fluctuations, and asymmetric operating states.

The superior control performance of the proposed method is predicated on the premise that accurate motor parameters are obtained in advance through offline parameter identification. However, in practice, motor parameters vary with operating conditions. Therefore, enhancing the global robustness of DPCC against motor parameter variations remains a critical topic for further research. Future work will explore the integration of advanced control algorithms, such as online parameter identification, disturbance observer-based compensation, and model-free predictive control.

Another potential concern is that during the reconfiguration transient, simultaneous conduction of all series and parallel switches may give rise to circulating currents caused by asymmetries including winding manufacturing deviations, inconsistent voltage drops across power devices, and mismatched drive signal delays. Mitigating this latent issue also constitutes a valuable research direction.

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Abbreviations

The following abbreviations are used in this manuscript:

SVPWM	Space Vector Pulse Width Modulation
PMSM	Permanent Magnet Synchronous Motor
ZVV	Zero-Voltage Vector
DPCC	Deadbeat Predictive Current Control
EVs	Electric Vehicles
ZC	Zero-current
PBP	Phase-by-Phase
ZVV-DPCC	The ZVV-based DPCC
FOC	Field-oriented Control

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